Coastal and Harbour Engineering

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Content

1. Introduction

This report presents the work carried out for the course *Coastal and Harbour Engineering*, part of the Master's program Sustainable Coastal and Ocean Engineering at Roma Tre University. The course was taught by Prof. Alessandro Romano, who provided clear explanations and very good guidance.

The aim of the report is to apply the theory from the lectures to practical coastal-engineering problems. First, short-term and long-term wave statistics are analyzed to understand the local wave climate. Then, linear wave theory is used to study wave propagation from offshore to the nearshore. Based on these results, a rubble mound breakwater is designed (including wave-breaking check, armour layer, filter layers and berm), and the stability of a vertical breakwater is checked for sliding and overturning using Sainflou and Goda methods. Finally, an extended exercise is carried out in which an old bathymetric chart is converted into a digital depth grid, a SWAN wave model is set up, and the resulting wave field is analyzed. In this way, the report shows the full chain from wave data analysis to the design and verification of coastal structures.

2. Exercises

2.1 Short-term analysis

Short-term analysis is used to describe a single sea state over a limited period. Instead of looking at wave statistics over months or years, we focus on a short snapshot, typically 10 to 30 minutes of continuous wave measurements. One single wave is not sufficient for statistical analysis, but using a record (over 30 minutes), too long may include changing conditions like storms, making the data less useful for representing a stable sea state.

The measurement data should be collected offshore, far from the influence of the seabed. To prevent the signal from being affected by the bottom, the water depth, h , should be at least about 100 meters. This ensures that measurements represent the actual sea surface elevation caused by waves, rather than shallow-water effects.

In these measurements, the free surface elevation is recorded on the vertical axis (y-axis) and the time on the horizontal axis (x-axis). The resulting data represents a complex mix of many different wave components: *wind-generated waves* (local short waves), *swell waves* (longer waves traveling from afar), and possible remnants of previous sea states. All these components superimpose to form the observed "sea state."

The main parameters we want to calculate from a short-term analysis are:

- The significant wave height (Hs)
- The significant wave period (Ts)

Both values provide a useful label for describing and comparing different sea states. Importantly, Hs and Ts always belong together as a pair: each sea state is defined by both its height and period.

Short-term wave analysis can be performed using two primary methods:

1. Zero-crossing analysis (time-domain):
   This method identifies when the surface elevation crosses the mean water level and measures the properties of each individual wave between consecutive upcrossings. Using this approach, the significant wave height (Hs) is determined as the mean of the highest one-third of all measured waves, written as $H(1/3)$. Similarly, the significant wave period (Ts) is computed as the average period of the same highest one-third of waves $T(1/3)$. Thus, Ts is directly paired with Hs based on the largest waves in the record.

2. Spectral analysis (frequency-domain):
   This method analyzes the energy content of the sea state using mathematical transforms such as the Fast Fourier Transform (FFT). Here, the significant wave height is computed as the spectral parameter Hmo, which is statistically equivalent to Hs but derived from the variance of the wave spectrum. The characteristic wave period is typically reported as the peak period Tp, representing the most energetic frequency in the spectrum.

In this exercise, we use the zero-crossing (time-domain) method.
By writing our code so that all upcrossing waves are located and measured, we can directly calculate both the significant wave height $H1/3$ and the significant wave period $T1/3$, focusing our analysis on the most relevant part of the sea state.

In this analysis, we focused on the highest one-third of all measured waves (leading to $H1/3$), but the same approach can be used for other fractions depending on the engineering need or convention. For example, you can calculate:

- $H1/5$: the mean height of the highest one-fifth of all waves,
- $H1/10$: the average of the highest 10% of waves,
- $H1/20$,
- $H1/100$, and so forth.

As you focus on increasingly rare and large waves (for example, $H1/100$), the calculated height rises, and when only the single largest wave in your measurement record is considered, this value becomes the maximum wave height ($Hmax$). Each of these statistical measures provides further insight for design, safety, or scientific purposes, by quantifying not just the “typical” but also the “extreme” wave conditions that can occur within a sea state.

These formulas for determining significant wave heights (such as $H1/3$, $H1/5$, $H1/10$, and so on) are valid because, in many real sea states, the distribution of individual wave heights closely follows a Rayleigh distribution. This statistical distribution accurately describes the probability of different wave heights occurring in a randomly mixed wave field, especially when the waves are the result of many components superimposed (as in short-term sea states).

$$Significant\ wave\ height = H_{1/3} = \frac{1}{N/3}\sum_{j=1}^{N/3} Hj$$

$$Significant\ wave\ period = T_{1/3} = \frac{1}{N/3}\sum_{j=1}^{N/3} T_0, j$$

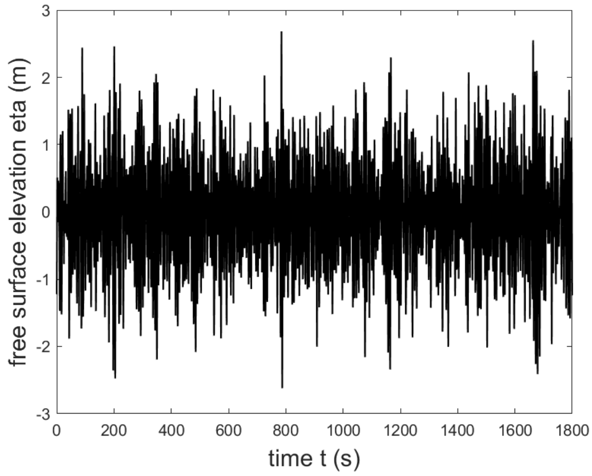

Figure 1: Visualization of the short-term sea state measurement. The x-axis represents time (t), and the y-axis shows the free surface elevation (eta).

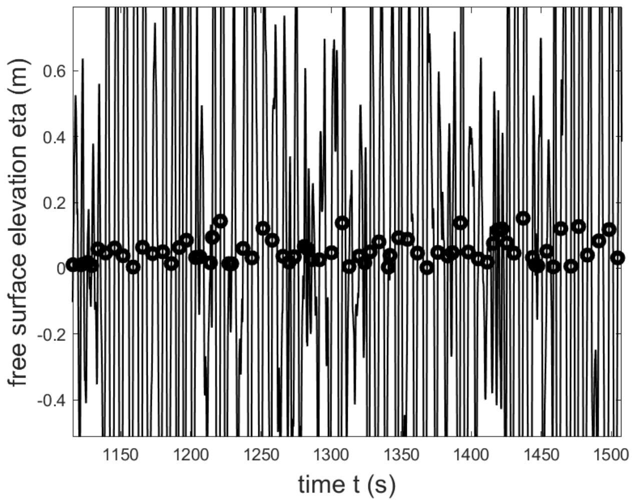

Figure 2: The same time series as Figure 1, now visualized together with the identified upcrossing waves. The upcrossings are marked, highlighting where individual wave calculations will be performed.

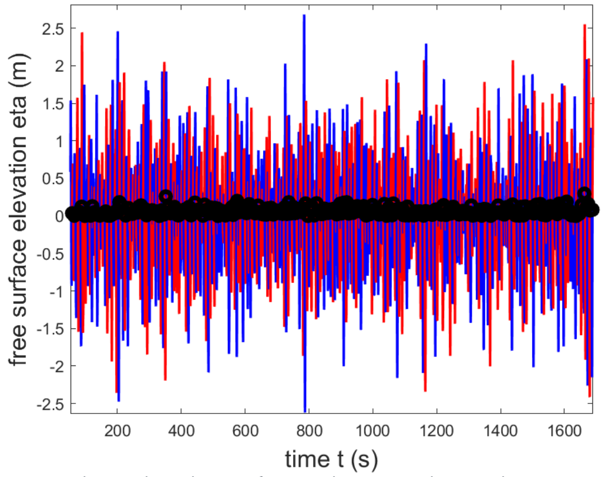

Figure 3: Enhanced visualization of the time series, with each individual wave segment shown in a different color. This clearly illustrates the separation of waves detected using the upcrossing method.

Summary of Short-Term Spectral Analysis

Short-term spectral analysis is used to describe the energy distribution of ocean waves within a relatively short time window (typically 10–30 minutes) during which the sea state is assumed to be stationary. Rather than analyzing individual waves, spectral analysis examines how the total energy of the sea surface is spread across different frequencies (or periods).

Step 1: Creating the Spectrum

Starting from a time series of sea surface elevations—measured at a fixed point offshore—we apply the Fast Fourier Transform (FFT) to convert the data from the time domain into the frequency domain. The resulting spectrum, usually denoted as S(f) or E(f) , shows how much energy is contained at each wave frequency. The x-axis represents frequency (Hz) or wave period (s), and the y-axis represents variance or energy density (m^2/Hz).

Step 2: Interpreting the Spectrum

Peaks in the spectrum correspond to the most energetic wave components. For example, a pronounced peak at a low frequency (which means a long wave period) indicates the presence of swell waves, these are large, long-traveling waves generated by distant storms. In contrast, a peak at a high frequency (short wave period) reflects local, wind-generated waves, often called wind waves. By analyzing where the peak occurs on the frequency axis, we can distinguish whether the sea state is dominated by long-period swell or by shorter, wind-driven waves.

Step 3: Determining Key Wave Parameters

- The significant wave height (H_{m0}) is found using the spectrum's total variance (m0), calculated as the area under the spectrum curve:

$$H_{m0} \approx 4\sqrt{m_0} \qquad m_0 = \int_0^{\infty} \int_{-\pi/2}^{\pi/2} S(f,\theta) df \, d\theta$$

- The peak period (Tp) is the wave period corresponding to the frequency where the spectrum reaches its maximum:

$$Tp = \frac{1}{fp}$$

  where fp is the frequency at the peak of the spectrum.

In summary:

Short-term spectral analysis provides a compact and robust description of the sea state, allowing to identify dominant wave conditions (through Tp), and overall wave energy (through H_{m0})Interpreting the location of spectrum peaks helps distinguish between swell-dominated and wind wave–dominated sea states, which is essential for coastal and offshore structure design.

2.2 Long-term statistics

As an engineer, designing safe coastal and offshore structures requires reliable knowledge of the sea state, not just its average behavior, but especially its rare extremes. Short-term wave analysis gives us detailed measurements for every 10–30 minute window (spectral in the exercise), but this is not enough for design purposes. For proper engineering, I need data covering long periods, ideally at least 40–50 years. The longer the data record, the more accurate my estimates of rare, extreme events become.

1. Data Preparation and Visualization
The first step is to collect as much short-term wave data (H_S or H_{mo} and Ts/Tp) as possible, aiming for at least 40–50 years of continuous records. More data is always better, since it gives a clearer picture of the wave climate and improves the reliability of statistical analysis, especially for estimating dangerous rare waves.

- I start by plotting H_S over time (years on the x-axis). This lets me see trends, seasonal changes, and visually pick out the largest waves.
- I create scatter plots of Tp versus H_S to distinguish between different types of waves (swell vs wind sea) and to spot any outliers.

A long dataset helps "see" both common and rare sea states, leading to safer design. Plots help me get a feel for the data and immediately detect any obvious abnormalities.

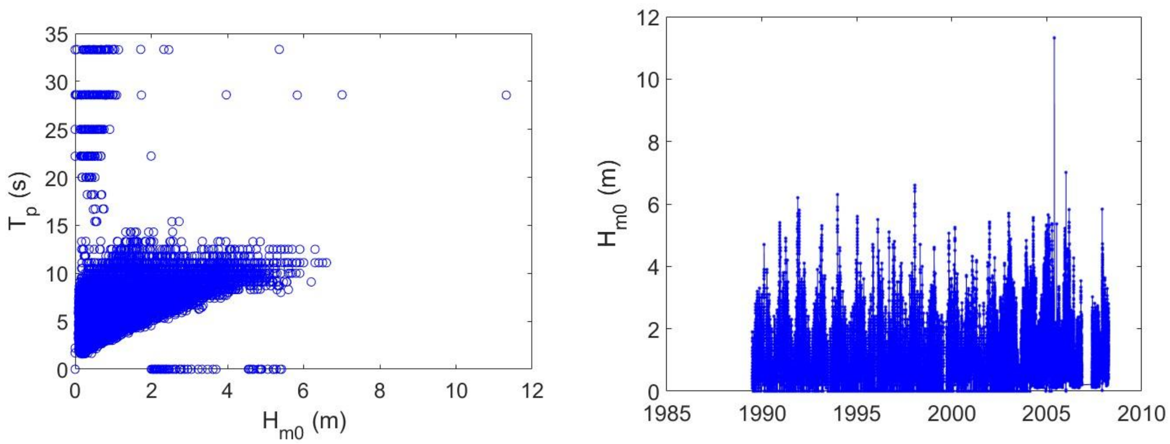

Figure 4: Data from spectral analysis, with errors

2. Data Cleaning and Directional Analysis

Once I've visualized the basic patterns, I clean the data to ensure further analysis is reliable.

- All measurements with impossible or physically implausible values (like Tp = 0 or very low H_S with very high Tp) are removed.
- I also generate a polar (directional) plot ("wave rose") to visualize which directions the strongest waves come from.

Clean data is very important, mistakes can make a design unsafe.
It's also important to know where waves usually come from. For coastlines and harbors, this helps us build structures in the best way to keep them safe and calm inside

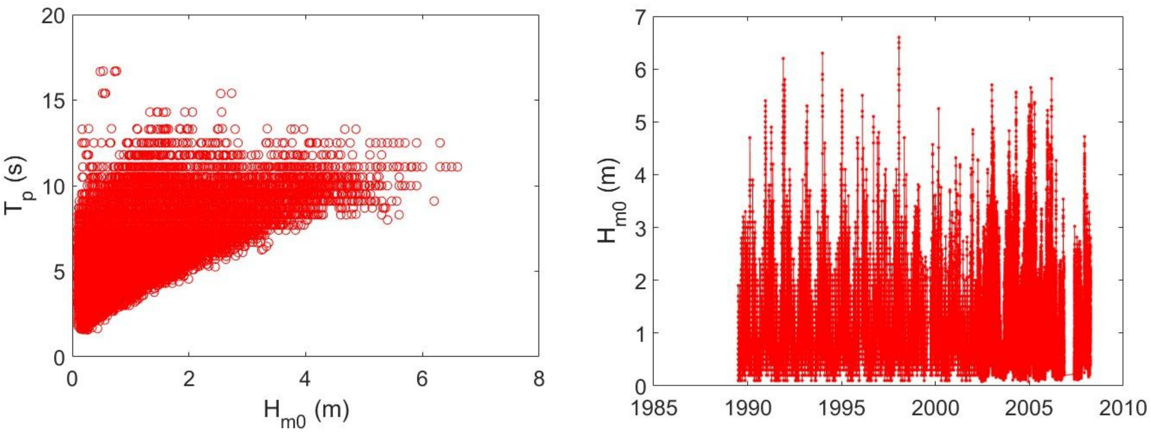

Figure 5: Cleaned data, wind waves.

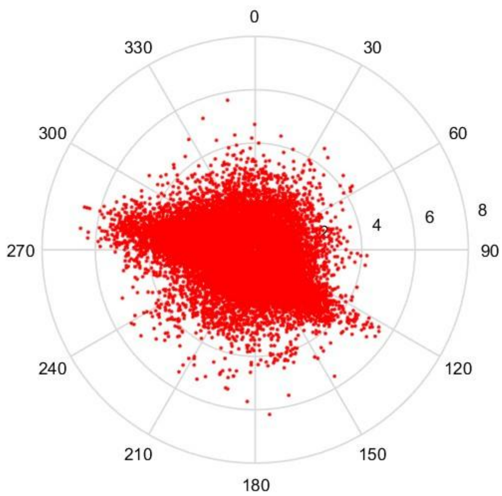

Figure 6: Polar plot, The waves with high Hs are coming from the West

3. Selecting Independent Extreme Events

Engineering design focuses on rare, high-impact events, not day-to-day averages. To study these extremes, I have two main tools:

3.1. Annual Maxima Method

I take the single highest Hs from each year. This gives me a set of annual maxima, with exactly one value per year.
Cautions:

1. If a major storm occurs at the end or start of a year, there's a risk of double-counting or splitting the event. This breaks the rule that each value must be independent.
2. This method is only reliable with long datasets (preferably 40–50 years or more). If the dataset is short, estimates of rare waves are highly uncertain.

3.2. Peaks Over Threshold (POT) Method

Instead of one value per year, I select all independent events where Hs exceeds a chosen high threshold.
The threshold must be chosen so we end up with about 3–4 extremes per year.
Cautions:

1. Big storms may produce many peaks above the threshold close together in time. These are *not* independent events.
2. To maintain independence, I only count one extreme value per storm period (typically using a separation of 48 hours).

Only independent, representative extreme events can be used to fit a statistical model that is meaningful for design.

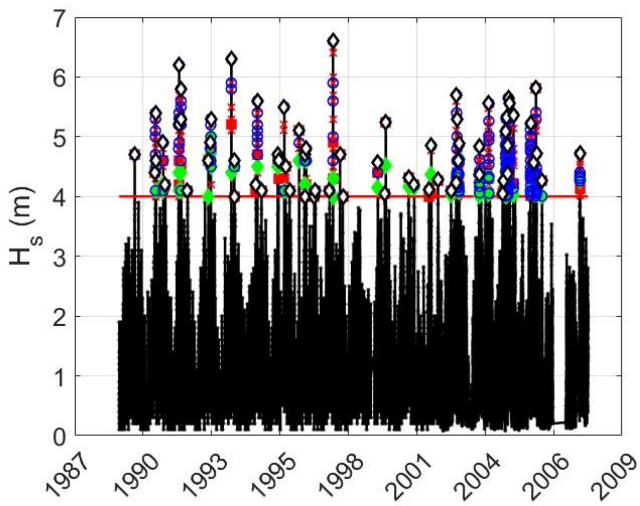

Figure 7: POT-method with the threshold(significant wave height 4 meters).

4. Fitting and Validating a Probability Distribution Function

With the right set of independent extreme values, I fit a probability distribution function (PDF), such as Gumbel, Weibull, or another extreme value distribution.
There are many possible PDFs, so I must check which one fits my data best.
I do this using a probability plot:

1. If my observed data points fall mostly on or below the fitted line (for example, the Gumbel line), then I know my PDF is adequately describing the extremes, and I can safely use its results for design parameters.
2. If many points are above the line, my PDF is underestimating the extremes. In this case, I have to try another PDF, because using an unfit distribution could result in unsafe designs.

The final design wave heights entirely depend on choosing and validating the right PDF. Using the wrong one can lead to under- or over-engineering (dangerous or wasteful).

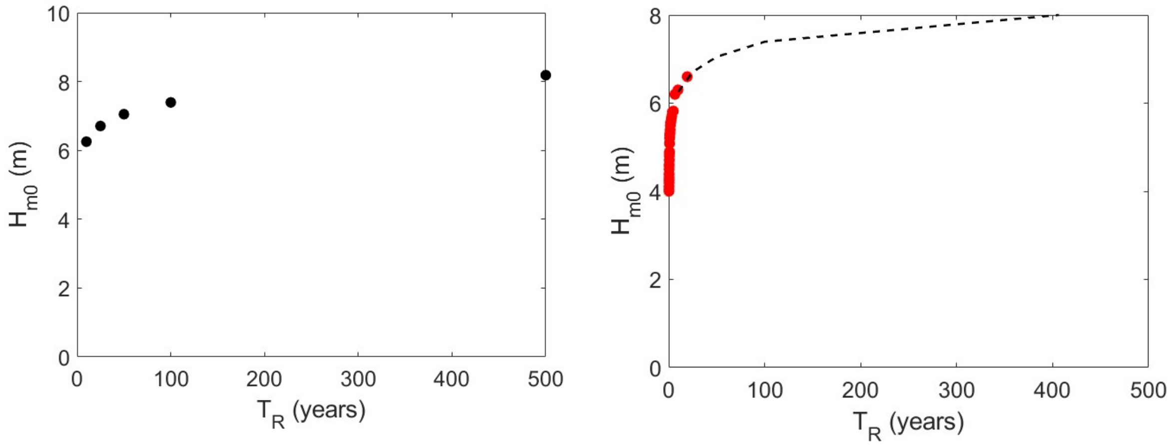

Figure 8: In black the Gumbel Probability Function, in red the observed extreme values from POT-Method.

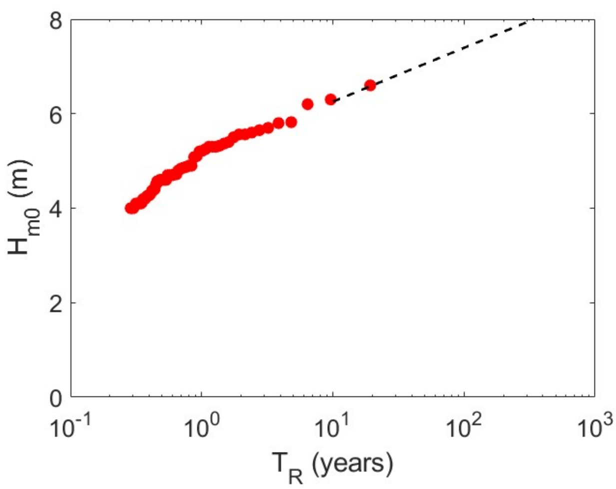

Figure 9: Logarithmic visualization, to analyze the plot more accurate. We can use the Gumbel because the red dots are on the Gumbel line.

5. Calculating Design Parameters and Probabilities

With a validated PDF, I can estimate:

- The return period wave height (for example, the 100-year wave, meaning the wave height that is statistically expected to be exceeded once every 100 years).
- Confidence intervals for these design values, showing how certain or uncertain these estimations are.
- Annual exceedance probability: Chance that the design wave is exceeded in any single year ($1/TR$).
- Lifetime/encounter probability: The probability that a structure will see at least one such wave over its whole lifespan (using: $P = 1 - (1 - 1/TR)^n$, where n is the lifetime in years).

These results transform raw data into meaningful numbers that directly inform design decisions, ensuring robustness and safety for the project.

A solid engineering approach to long-term wave analysis requires a minimum of 40–50 years of high-quality, short-term wave data. I visualize and clean this data, carefully select independent extreme events (using annual maxima or POT), fit and validate the right probability distribution, and finally calculate critical design wave heights and their likelihood.

Each step, especially fitting and validating the PDF, is crucial: if the data fall on or below the fitted curve, the design values are safe to use. If not, I must find a better fitting distribution, as underestimating extremes could directly compromise safety.
By following this process, I ensure all design choices for coastal and offshore infrastructure are based on proven and reliable statistical analysis.

2.3 Linear Wave Theory: Part 1

Dispersion Relationship

The dispersion relationship is a fundamental equation in the linear theory of water waves, describing how wave frequency, wavenumber, and water depth are interconnected. Starting from the boundary conditions applied at the free surface, and implementing two time derivatives and one spatial derivative, we arrive at an explicit relationship that links the angular frequency (ω), the wavenumber (k), and the local water depth (h):

$$\omega^2 = gktanh\,(kh)$$

where:

- Omega ω is the angular frequency, with
- g is gravitational acceleration (9.81 m/s²),
- k is the wavenumber, defined as $k = \frac{2\pi}{L}$, with L as the wavelength,
- h is the local water depth,
- tanh(kh) is the hyperbolic tangent function, which quantifies the effect of depth on wave dispersion.

This relationship is transcendental in k , which means that k cannot be calculated directly with a simple formula if you only know omega and h . Instead, you must use numerical or approximate methods to find k.

T	h	L	c	cg	h/L	DepthClass
7	100	76.5042	10.9292	5.4646	1.3071	"Deep Water "
7	10	59.3750	8.4821	6.4354	0.1684	"Intermediate "
7	1	21.6211	3.0887	3.0051	0.0463	"Shallow Water "
4	100	24.9810	6.2452	3.1226	4.0030	"Deep Water "
4	10	24.5212	6.1303	3.2520	0.4078	"Intermediate"
4	1	11.9874	2.9969	2.7539	0.0834	"Intermediate"
600	1000	5.9316e+04	98.8606	98.4928	0.0169	"Shallow Water"
600	100	1.8789e+04	31.3151	31.3034	0.0053	"Shallow Water"

Table 1 : Wave data quantities

Where:

The wave period (T) and the water depth (h) are given. Using the dispersion relationship, we can determine the wavelength (L), the phase celerity (c), and the group celerity (c_g). At the end, we classify whether the wave is in deep, intermediate, or shallow water using the ratio h/L.

Physically, the dispersion relationship shows that both the frequency and the wavelength of a wave depend on the water depth.

In deep water kh approaches ∞, which simplifies the dispersion relation to:

$$\omega^2 = gk$$

The phase celerity simplifies to:

$$c = \sqrt{\frac{g}{k}} = \frac{gT}{2\pi}$$

This shows that in deep water, longer waves (larger T) travel faster than shorter waves. The system is dispersive because wave speed depends on wavelength.

In shallow water kh approaches 0, what results in the dispersion relation in shallow water:

$$\omega = k\sqrt{gh}$$

so the phase celerity becomes:

$$c = \sqrt{gh}$$

Here, the wave speed depends only on water depth, not on wave period or wavelength. Therefore, in shallow water, all waves travel at the same speed—this is a non-dispersive system.

To classify the wave regime:
Deep water: $h/L > 1/2$
Intermediate water: $1/20 < h/L < 1/2$
Shallow water: $h/L < 1/20$

Whether the water is deep or shallow compared to the wavelength determines if the waves are dispersive or not. This theoretical framework is essential for understanding and predicting wave behavior in coastal and offshore engineering.

In practical wave calculations, it is common to first compute the deep water values for wavelength (L_0) and wavenumber (k_0) using the simple formulas $L_0 = \frac{gT^2}{2\pi}$ and $k_0 = \frac{2\pi}{L_0}$. These formulas can be applied directly because, in deep water, the effect of the seabed is negligible and the relationship between wave period and wavelength is straightforward.

However, for intermediate or shallow water conditions, the influence of the bottom becomes significant and the actual wavelength and wavenumber differ from the deep water values. In these cases, the deep water numbers (L_0 and k_0) serve as useful starting points. We then use the full dispersion relationship, often solved iteratively, to find the real wavelength (L) and wavenumber (k) that apply for the given water depth and wave period. This stepwise approach streamlines the calculation process and ensures accuracy when the seabed plays a role in wave propagation.

$$L \cong L_0 tanh\,[\,sinh\,(\sqrt{k_0 h}\,)]$$

Particle velocity

The particle velocities beneath a wave can be described using linear wave theory by introducing the velocity potential ϕ (x,z,t). The horizontal and vertical particle velocities then follow from

$$u(x,z,t) = \frac{\partial x}{\partial \phi} \qquad w(x,z,t) = \frac{\partial z}{\partial \phi},$$

where u is the horizontal velocity component and w the vertical component.
For a monochromatic wave with period T = 10 ,s, wave height H = 3.0 ,m and constant water depth h = 30 m, the vertical profiles of u(z) can be plotted at different phases of the wave.

In the crest phase, the profile is taken when the free surface elevation η reaches its maximum; in the trough phase, the profile corresponds to the minimum surface level. These profiles show how the horizontal velocity decreases with depth and how it changes sign between crest and trough.

Integrating the velocities in time gives the particle trajectories. In linear theory, these paths are ellipses that can be written schematically as

$$\frac{(X_p(t) - X_0)^2}{\alpha} + \frac{(Z_p(t) - Z_0)^2}{\beta} = 1,$$

where (X0, Z0) is the centre of the ellipse, (Xp(t) ,Zp(t)) is the particle position, and α and β represent the squared maximum horizontal and vertical excursions, respectively.

- In deep water ($kh \rightarrow \infty$), the horizontal and vertical excursions are equal and decay exponentially with depth:

$$\beta = \alpha = a\,\mathrm{e}^{-kz},$$

with a = H/2 the wave amplitude and k the wave number. The trajectories are nearly circular near the surface and become very small at greater depth; particles “do not feel” the bottom.

- In shallow water ($kh \rightarrow 0$), the motion becomes more horizontal and the influence of the bed is important. The horizontal excursion remains almost depth-independent, while the vertical excursion is much smaller near the bed. As a result, the trajectories are strongly elliptical and the whole water column is affected by the wave motion.

Pressure beneath water waves
At any location below the surface, the total pressure is the sum of the static weight of the water (hydrostatic pressure) and the oscillatory pressure created by the waves (dynamic or excess pressure).

Hydrostatic pressure
The hydrostatic pressure at a certain depth z (measured upward with z = 0 at the still water level) is given by

$$p^{I} = \rho g(-z),$$

Where:

- ρ is the water density,
- g is the gravitational acceleration,
- z is the vertical coordinate (negative below the still water level).

This term represents the pressure due to the static water column.

Excess (dynamic) pressure in linear wave theory
Superimposed on the hydrostatic term is the excess pressure due to wave motion. In linear wave theory this can be written as

$$p(x,z,t) = \rho g a \,\frac{\cosh\left(K(z+h)\right)}{\cosh\left(Kh\right)} \cos\left(Kx - \omega t\right),$$

where

- a = H/2 is the wave amplitude,
- k = 2π/L is the wave number,
- L is the wavelength,
- h is the water depth,
- ω = 2π/T is the angular frequency.

The total pressure under a wave is therefore

$$p(x,z,t) = p^{I}(z) + p^{+}(x,z,t).$$

The behaviour of the excess pressure with depth depends strongly on the relative depth kh.

Deep water

In deep water ($kh \rightarrow \infty$), the depth is much greater than the wavelength. In that case the excess pressure decays approximately exponentially with depth:

$$p^{+}(z) \propto \rho g a\, \mathrm{e}^{-kz}.$$

This means that the wave-induced pressure variations are strong only near the free surface and rapidly become negligible further down. Consequently, in deep water only structures or features close to the surface experience significant dynamic pressure.

Shallow water

In shallow water ($kh \rightarrow 0$), the depth is small compared to the wavelength and the excess pressure varies only weakly with depth. In the limit, the dynamic pressure can be considered nearly constant over the depth:

$$P^{+} \approx \pm \rho g a,$$

with a positive value under a crest and a negative value under a trough. In shallow water, the entire water column, including the seabed, "feels" almost the full effect of the passing wave crests and troughs.

Example calculations

The excess pressure due to waves can be evaluated at different depths and phases of the wave motion. For a monochromatic wave with period T = 6.5s, wave height H = 2.0m and water depth h = 8m:

- During the crest phase at z = -6,m, the excess pressure p+ is 0.7804,Pa.
- During the trough phase at z = -4,m, the excess pressure p+ is -0.8228,Pa.

These values illustrate how the dynamic pressure depends both on depth and on the wave phase (crest versus trough).

Finally, for a monochromatic wave with period T = 5.0s, wave height H = 1.0m and water depth h = 100m, the total pressure profile can be obtained by summing the excess pressure and the hydrostatic pressure at each depth:

$$p(z,t) = (-\rho g z) + p^{+}(z,t).$$

By plotting this total pressure as a function of depth at different phases of the wave, the combined effect of hydrostatic and dynamic contributions becomes clear. This is consistent with the pressure profiles shown in the corresponding figures.

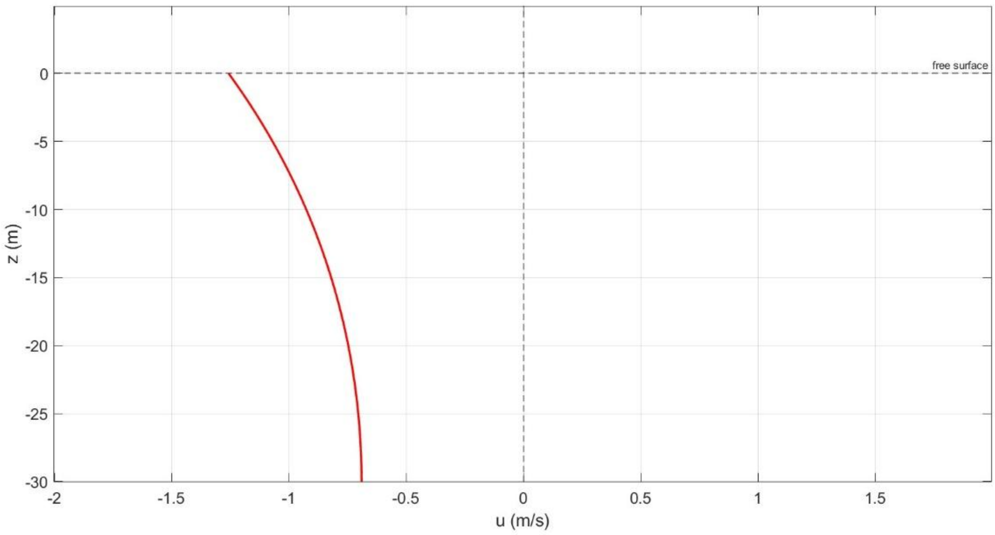

Figure 10: Vertical profile of the horizontal velocity u(z) as a function of depth z during the trough phase of the wave.

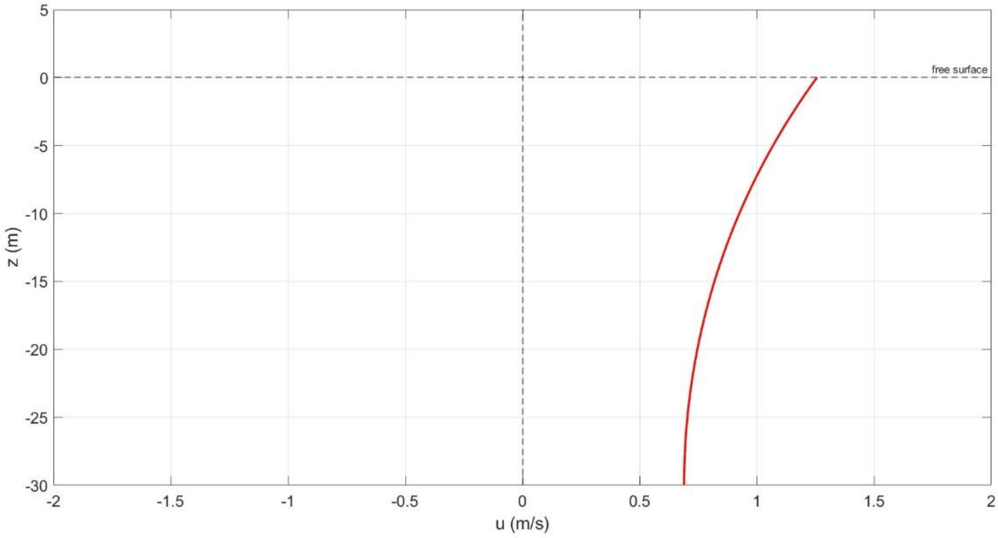

Figure 11: Vertical profile of the horizontal velocity u(z) as a function of depth z during the crest phase of the wave.

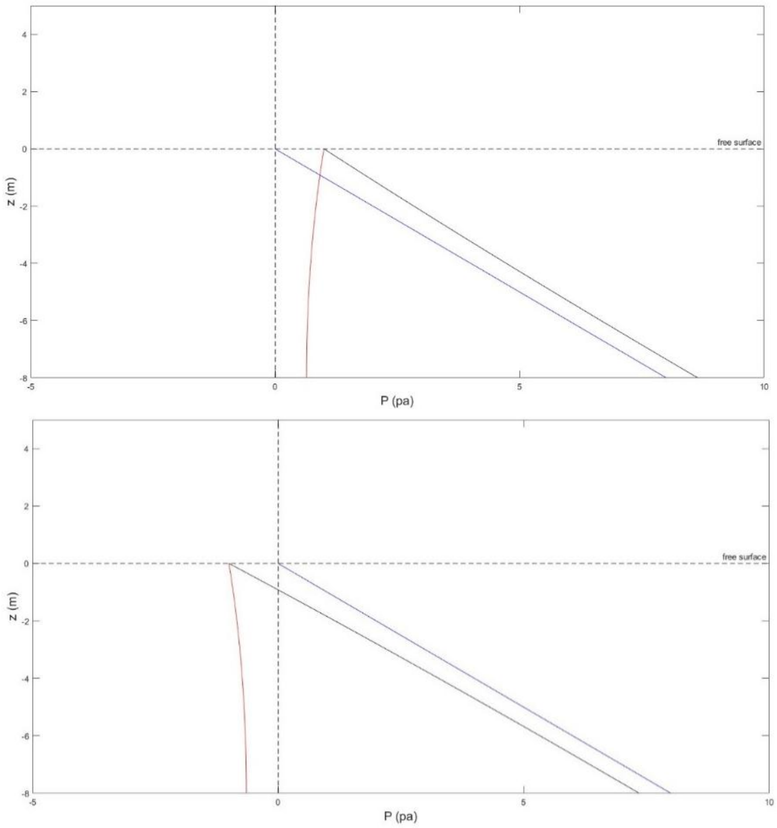

Figure 12 & 13: In these figures, the excess pressure p+ is indicated in red. Figure 12 shows the excess pressure below the wave crest, while Figure 13 shows the excess pressure below the wave trough.

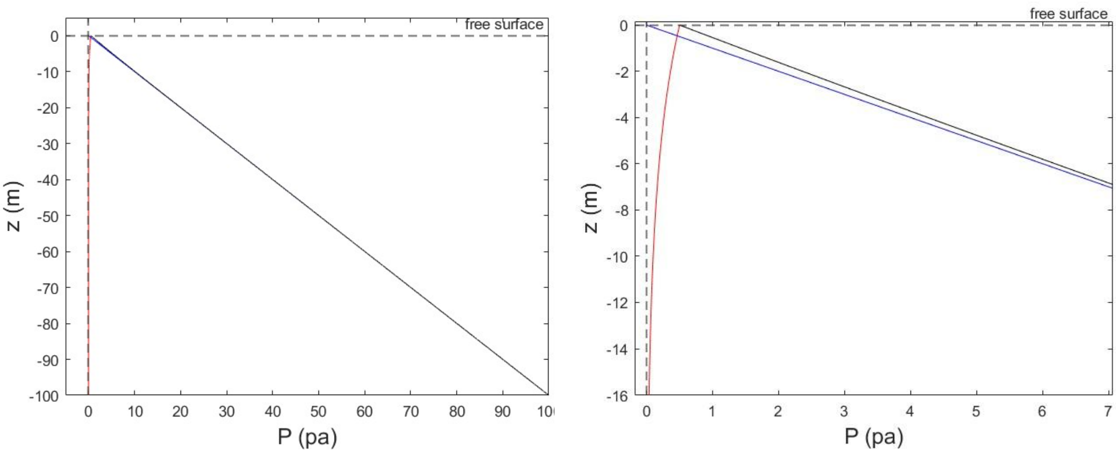

Figure 13 & 14 : These figures show the hydrostatic pressure in blue, the excess pressure in red, and the total pressure in black, all during the crest phase of the wave. Figure 14 is a zoomed-in view of Figure 13.

2.4 Linear Wave Theory: Part 2 (Wave propagation from offshore to shore)

In coastal engineering practice, wave data is typically collected offshore, for example using a wave buoy. Such instruments provide us with the key wave parameters offshore: the significant wave height (H_0), wave period (T_0), and wave direction (ϑ)(relative to the shoreline). However, as engineers, we are often most interested in the behavior and impacts of waves closer to the shoreline, where they affect coastal structures and processes. Therefore, it is necessary to propagate the offshore wave data towards the nearshore area of interest.

During this propagation, two main physical phenomena affect the waves:

1. Shoaling, the change in wave characteristics due to decreasing water depth.
2. Refraction, the change in wave direction due to depth-dependent speed.

To analyze shoaling, we first consider wave propagation perpendicular to the shore (with zero approach angle). As waves travel from deeper to shallower water, the water depth h decreases, leading to a reduction in wavelength (L) and wave celerity (c). Conservation of energy principles dictate that, as the wave slows down and is compressed, the wave height H must increase.

For this study, we assume:

- A straight shoreline,
- A uniformly sloping ("plain") beach with a mild gradient,
- No energy loss due to wave breaking, bottom friction, wind or other sink and source terms.
- The beach profile is simple and smooth (mildly varying depth).

From the conservation of energy:

$$\frac{\partial E}{\partial t} + \nabla \cdot (E\vec{c}_g) = 0$$

If there is no breaking (no energy loss), and the system is steady $\frac{\partial E}{\partial t} = 0$, the energy flux in the direction perpendicular to the coast is constant.
Because we are propagating the wave along one principal direction (from offshore to onshore), the flux in the alongshore direction is zero, and we neglect any source or sink terms. Thus:

$$E_1 c_{g1} = E_0 c_{g0}$$

where:

- $E = \frac{1}{8}\rho g H^2$ is the wave energy.
- c_g is the group velocity.

Subscripts 0 and 1 refer to offshore and a nearshore position, respectively.
From this, the shoaling equation can be derived:

$$H_1 = \mathrm{H}_0 \sqrt{\frac{c_{g0}}{c_{g1}}} \sqrt{\frac{\cos\theta_0}{\cos\theta_1}}$$

where:

- The Shoaling coefficient Ks = $\sqrt{\frac{c_{g0}}{c_{g1}}}$
- The Refraction coefficient Kr = $\sqrt{\frac{\cos\theta_0}{\cos\theta_1}}$

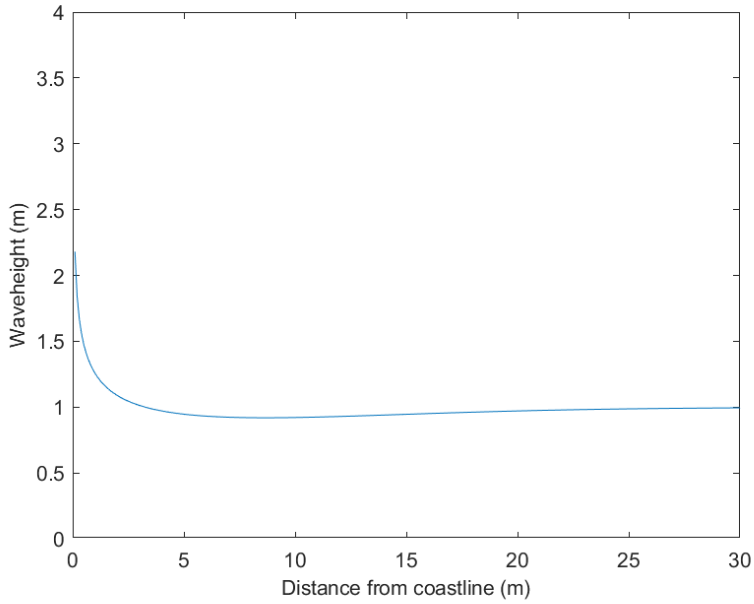

Figure 15: Shoaling

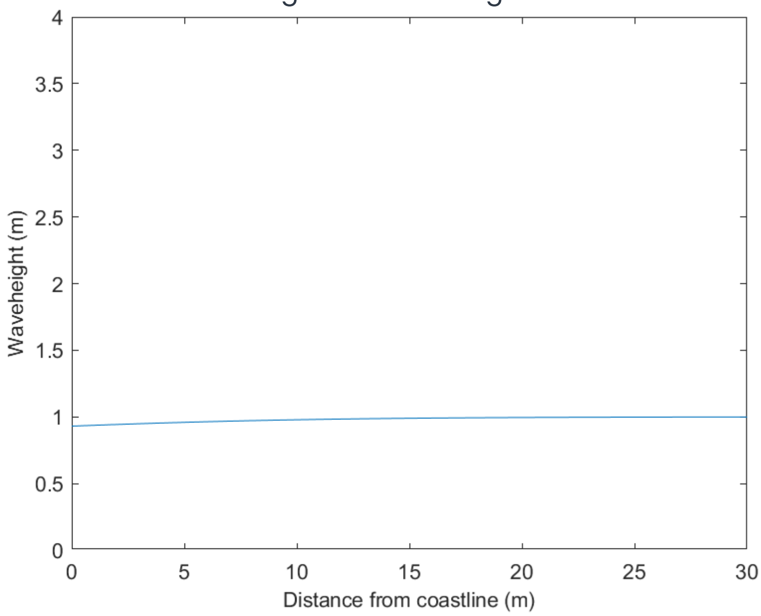

Figure 16: Refraction

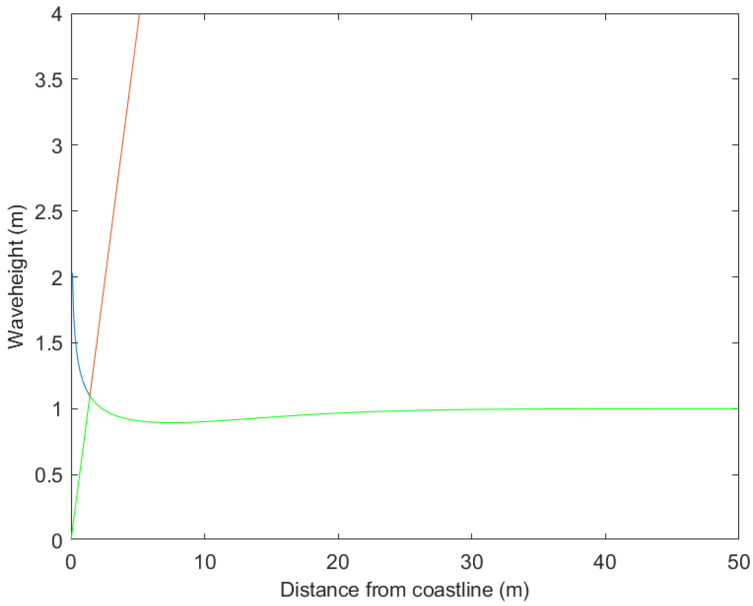

Figure 17: Breaking line in red (Hmax=0.78hb). Refraction + shoaling in blue. Green the real behavior.

This formula allows us to predict the change in wave height (H) as the wave approaches the coast and the water becomes shallower, before taking into account any effects from wave refraction or breaking.(Figure 15)

In the resulting figure, we observe that, as the wave moves from deep to shallower water, the wave height actually decreases by about 8–9%. This is due to the adjustment of group velocity and energy flux in the still relatively deep water. As the wave continues to propagate towards the shore and the depth becomes much smaller, we see the wave height sharply increase towards infinity as predicted by linear theory (if no breaking is considered). Of course, in reality, wave breaking will limit the maximum wave height near the shoreline.

2.5 Rubble Mound Breakwater Design

2.5.1 Wave breaking check

Step 1 is to check whether the waves are breaking at the toe of the rubble mound breakwater. We do this by comparing the maximum wave steepness from the Miche criterion with the actual wave steepness at the toe.

If the actual wave steepness is higher than the Miche criterion, the waves are breaking and the rubble mound breakwater must be designed heavier and stronger. If the actual wave steepness is lower, the waves are non-breaking and the required armour can be less heavy.

Hypothesis:

- Water depth at the toe of the structure h = 12.0 m.
- Constant sea level (i.e. no astronomical or meteorological tides are considered).
- Design wave height (evaluated at the toe of the structure) Hs = 6.0 m .
- Related wave period Tp = 12.0 s.
- Wave fronts are parallel to the main line of the structure.

With these hypothesis we can calculate the wavelength.
First, the deep-water wavelength is calculated as:

$$L_{deepwater} = \frac{g\,T_p^2}{2\,\pi}$$

Then the deep-water wave number is:

$$k_{deepwater} = \frac{2\,\pi}{L_{deepwater}}$$

Finally, the wavelength at the local water depth is computed as:

$$L_{toe} = L_{deepwater} * \tanh\left(\sinh\left(\sqrt{k_{deepwater} * h}\right)\right)$$

First of all we calculate the maximum steepness with the next formula. (Miche breaking conditions)

$$\varepsilon_{max} = 0.142\tanh\left(\frac{2\pi h}{L}\right)$$

Where:

- L is the wavelength at the water depth h
- ε_{max} : Maximum (limiting) wave steepness, i.e. the largest wave steepness allowed before breaking is expected according to Miche's criterion.
- h is the Water depth, the vertical distance from the still water level to the toe of the rubble mound breakwater.

We then compare this maximum steepness with the actual wave steepness, which is given by the ratio of the significant wave height to the wavelength.

$$\varepsilon_{\text{actual}} = \frac{H_s}{L}$$

If $\varepsilon_{\text{actual}}$ is larger than ε_{max} the waves are expected to break and the rubble mound breakwater must be designed heavier and stronger. The calculated maximum wave steepness for our study case, based on Miche's criterion, is:

$$\varepsilon_{max} \approx 0.078$$

The actual wave steepness is:

$$\varepsilon_{\text{actual}} \approx 0.049$$

Since the actual wave steepness is lower than the maximum allowable steepness, the waves at the toe of the rubble mound breakwater are classified as non-breaking. Therefore, non-breaking wave conditions are assumed in the design steps.

2.5.2 Design of the armour layer and filter layers

Now that we know the waves are non-breaking at the toe, we can start designing the armour layer.

For the armour layer design, the following assumptions are made:

- Slope of the structure: (3/4) (vertical/horizontal)
- Type of armour unit: artificial concrete unit tetrapod
- Specific weight of the tetrapod material $\gamma_s : 2.4\ t/m^3$

To estimate the required weight of the armour units, the Hudson formula is applied.

$$P = \frac{\gamma_S\ H_S^3}{\Delta^3\ K_D\ cotg(\alpha)}$$

where:

- γ_s : Specific weight of the tetrapod material.
- $\mathrm{H_s}$: Design wave height (evaluated at the toe of the structure)
- $\Delta : (\frac{\gamma_s}{\gamma_w}) - 1$ specific weight of tetrapod divided by specific weight of salty water
- $\mathrm{K_D}$: Given in Tabel 1
- g : the gravitational acceleration.
- α : angle of the armour slope.

In our study case, the required armour unit weight of the tetrapods, calculated using Hudson's formula, is:

$$P \approx 20.65\ \mathrm{t}$$

This value represents the design weight of a single tetrapod in the primary armour layer of the rubble mound breakwater.

Armour Block	N. of layers	Coeff K_Δ	Porosity %	Current Section K_D breaking wave	Current Section K_D non-breaking wave	Head K_D breaking wave	Head K_D non-breaking wave	slope cotg α
Natural stone (sharp edges)	1	-	-	(a)	2,90	(a)	2,30	1,50 ÷ 3,00
Natural stone (smooth edges)	2	-	38	1,20	2,40	1,20	1,90	1,50 ÷ 3,00
Natural stone (sharp edges)	2	1,00	37	2,00	4,00	1,90 - 1,60 -1,30	3,20 - 2,80 - 2,30	1,50 - 2,00 - 3,00
Natural stone (sharp edges)	≥3	1,00	40	2,20	4,50	2,10	4,20	1,50 ÷ 3,00
Parallelepiped	2	1,10	-	5,00	6,00	3,00	3,50	2,50 ÷ 3,00
Antifer Cube	2	1,10	45	6,50	7,50	-	5,00	1,50 ÷ 2,50
Tetrapode	2	1,04	50	7,00	8,00	5,00 - 4,50 - 3,50	6,00 - 5,50 - 4,00	1,50 - 2,00 - 3,00
Dolos	2	0,94	56	15,00(b)	31,00(b)	8,00 - 7,00	16,00 - 14,00	2,00 - 3,00
Acropode II - Ecopodo	1	-	56	16,00(c)	16,00	12,30 (c)	12,30	1,33
Core-Loc®	1	-	60	16,00(c)	16,00	13,00 (c)	13,00	1,33

Table 1:

Now that the required weight of the tetrapod has been determined, the thickness of the armour layer can be calculated using the following formula:

$$r = n \cdot K_{\Delta} \left(\frac{P}{\gamma_S}\right)^{\frac{1}{3}}$$

Where:

- r : thickness of the armour layer in meters.
- n : number of armour units in the radial direction.
- K_{Δ} : dimensionless coefficient that accounts for the shape and packing of the armour units, found in table 1.
- P is the weight of a single tetrapod in tons.
- γ_s : Specific weight of the tetrapod material.

In our study case, the required armour layer thickness, this thickness corresponds to a two-layer tetrapod armour.

$$r \approx 4.26 \text{ m}$$

In our study case, the width of the armour layer was determined using the following relation:

$$b = 1.5 * r$$

Where:

r : thickness of the armour layer.

Once the dimensions of the armour layer were established, the filter layers underneath the armour were designed. Their thicknesses were calculated based on reduced unit weights, taken as fractions of the armour unit weight:

$$\mathrm{P}_1 = \frac{P}{10} \qquad P_2 = \frac{P}{100}$$

The corresponding layer thicknesses of the first and second filter layers were then computed using:

$$r_1 = n \cdot K_{\Delta} \left(\frac{P_1}{\gamma_S}\right)^{\frac{1}{3}} \qquad r_2 = n \cdot K_{\Delta} \left(\frac{P_2}{\gamma_S}\right)^{\frac{1}{3}}$$

Where:

- r_1 r_2 : thicknesses of the first and second filter layers.
- K_{Δ} : shape and packing coefficient for the filter material, (study case 1.15.)
- n is the number of layers, (study case 2.)
- γ_S : unit weight of the rock.
- P : the armour unit weight.

Based on the armour unit weight $P = 20.65\ t$, the characteristic unit weights of the first and second filter layers are:

$$P_1 \approx 2.07 \text{ t}, \quad P_2 \approx 0.21 \text{ t}$$

Using these values in the layer-thickness formula, the resulting filter layer thicknesses are:

$$r_1 \approx 2.10 \text{ m}, \qquad r_2 \approx 0.98 \text{ m}$$

2.5.3 Design of berm

At the toe of the rubble mound breakwater, a berm made of natural rock is provided. The stones have a specific weight:

$$\gamma_s : 2.4\ t/m^3$$

The berm is designed for a significant wave height:

$$Hs\ =\ 6.0\ m$$

and an incipient damage level, corresponding to a damage parameter:

$$N_{od}\ =\ 0.5$$

Determination of stone size and weight
First, the nominal median stone size D_{n50} of the berm stones is determined using the Gerding formula:

$$\frac{Hs}{\Delta \cdot D_{n50}} = \left(0.24\ \frac{d}{D_{n50}} + 1.6\right)\ N_{od}^{0.15}$$

Where:

- Hs : the significant wave height at the berm,
- Δ : the relative submerged density of the stones,
- D_{n50} : the nominal median diameter of the berm stones,
- d : the height of the berm crown below mean sea level (MSL),
- $N_{od}^{0.15}$: the damage parameter.

From the resulting D_{n50}, the mean stone weight is obtained as:

$$\bar{P} =\ D_{n50}\ *\ \gamma_s$$

Characteristic berm stone weight of:

$$P_b \approx 6.8\ t$$

The width of the berm crown is chosen such that at least three stones can be placed across the berm. This is done using:

$$b_{berm} = n \cdot K_\Delta \left(\frac{P_b}{\gamma_S}\right)^{\frac{1}{3}}$$

With K_Δ = 1.15 and n =3, we obtain $b_{berm}\ \approx 4.69\ m$

In addition, the berm elevation d is selected such that the recommended geometric criterion:

$$0.4 < \frac{d}{h} < 0.9$$

Where:

- d : the height of the crown of the berm below mean sea level.
- h : water depth at toe of the berm.

In our study case: 0.69. So $0.4 < 0.69 < 0.9$, is approved.

2.5.4 Design optimisation, Comparison of armour layers

Aspect	**Tetrapods**	**Accropodes**	**Natural rock armour**
Stability	Good interlocking; relatively high stability coefficient. Normally two layers leads to thicker armour. Sensitive to breakage if badly placed or overloaded.	Optimized for hydraulic stability and interlocking. Single-layer system leads to thinner armour and less concrete. Highest stability if placement rules are followed carefully.	Lower stability coefficient; needs larger stones and/or thicker layers. Very robust and less sensitive to failure of individual units.
Environment	Concrete leads to higher Co_2 per volume than rock; more artificial look and stronger visual impact. Shapes can create microhabitats for marine life over time.	Same as tetrapod's: concrete with higher CO_2 footprint and artificial appearance; also offers microhabitats once colonised.	Seen as more natural and visually better integrated. Quarrying and transport still have impacts, but embodied energy per mass is usually lower than for precast concrete.
Costs	Higher unit cost than rock; need molds, casting and storage. Better stability can reduce total volume. Overall medium to high, depending on local rock and labor.	Patented system with license and strict quality control. Highest unit cost, but single layer reduces total concrete volume. Often cost-effective for large projects, relatively expensive for small ones.	Usually cheapest per ton if local quarries exist. Larger required volumes and transport can make total cost similar to or higher than concrete units, especially over long distances.
Construction	Requires casting yard, curing time and storage. Placement with cranes in two random layers → slower and more sensitive to placement errors, difficult in rough seas.	Placement highly controlled, often with GPS or grids; demands skilled operators and good site management. Once organised, single-layer placement can be relatively fast.	Can be placed with standard earthmoving equipment. No moulds or curing. Method is simple and flexible; repairs are easy, but good grading and quality control are needed for interlocking and layer thickness.

- **Accropodes**: highest hydraulic efficiency and most compact section, but licensed, concrete-intensive and execution-sensitive.
- **Tetrapods**: proven concept with good stability, but need two layers and more concrete.
- **Natural rock**: usually the most natural and often cheapest per tonne, but requires more volume and thicker layers for the same safety level.

2.5.5 Technical drawing of Rubble Mound Breakwater

2.6 Vertical Breakwater Check

In this part of the report, the stability of a vertical breakwater is checked. The analysis is divided into three main steps: Input data and geometry, force calculation and a sliding stability check of the vertical breakwater is performed at the end of the chapter.

2.6.1 Input data and geometry

For the vertical breakwater, the following geometric, hydraulic and geotechnical data are given (see Figure 18):

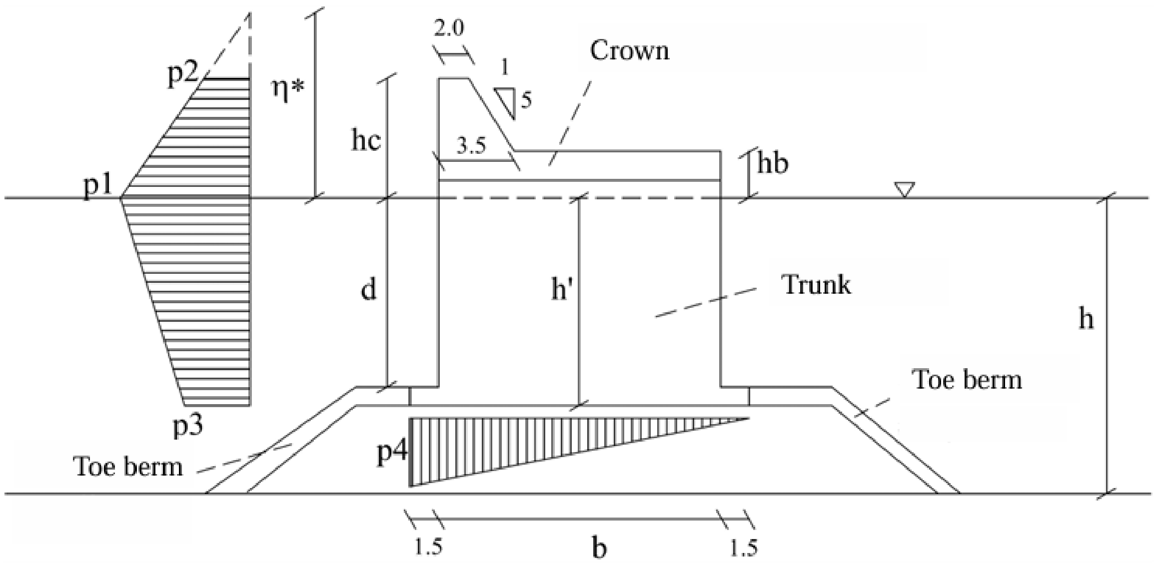

Figure 18: Vertical breakwater

Geometry:

- Water depth at the trunk:
  $h = 20\ m$

- Distance from still water level to caisson base (embankment height):
  $d = 15\ m$

- Effective height of the caisson trunk:
  $h' = 16\ m$

- Width of the caisson base (trunk width):
  $b = 16\ m$

- Freeboard of the crown above still water level:
  $hc = 10\ m$

- Height of the crown deck above the MSL
  $hb = 2.5\ m$

Wave conditions:

- Significant wave height:
  $Hs = 5\ m$

- Peak wave period:
  $Tp = 12\ s$

Soil–structure interaction and safety:

- Static friction coefficient at the base:
  $\mu = 0.6$

- Required sliding safety factor:
  $Cs = 1.4$

  Unit weights:
- Unit weight of concrete (trunk):
  $\gamma_{Tr} = 2.1 \cdot 9.81\ kN/m3$

- Unit weight of concrete (crown):
  $\gamma_{Cr} = 2.3 \cdot 9.81\ kN/m$

- Unit weight of seawater:
  $\gamma_{W} = 1.3 \cdot 9.81\ kN/m3$

These data form the basis for the next step, where we will calculate the forces on the vertical breakwater (without waves and with waves, using Sainflou and Goda for the horizontal wave loads) and then perform the sliding check.

2.6.2 Force calculations

2.6.2.1 Forces without waves

Before considering wave loading, the static forces acting on the vertical breakwater are determined. These are the weight force of the caisson (trunk + crown) and the buoyancy force.

Weight force:
The total weight of the structure is composed of:

- the weight of the trunk:

$$P_{\mathrm{Tr}} = b \ \cdot (h' + 1)\, \gamma_{\mathrm{Tr}}$$

- the weight of the crown:

$$P_{\mathrm{Cr}} = \left[(h_b - 1) \cdot b + (h_c - h_b) \cdot \frac{3.5 + 2.0}{2}\right] \gamma_{\mathrm{Cr}}$$

The total weight force of the caisson then becomes:

$$P = P_{\mathrm{Tr}} + P_{\mathrm{Cr}}$$

Buoyancy force:
The buoyancy force acting on the base of the caisson is calculated as:

$$F_g = b \cdot h' \cdot \gamma_w$$

In our study case, these expressions give a total weight force of:

$$F_g \approx 6.61 \times 10^3 \text{ kN}$$

and a buoyancy force of:

$$F_b \approx 2.59 \times 10^3 \text{ kN.}$$

2.6.2.2 Forces with waves and sliding check

Forces with waves

When waves are present, additional horizontal wave forces act on the vertical breakwater. These forces are evaluated using two methods: Sainflou and Goda. Before Sainflou's method can be applied, it is necessary to verify that the waves at the structure are non-breaking.

1) **Sainflou's theory**

Sainflou's theory can be applied only if waves do not break in front of the vertical wall. A practical rule of thumb states that wave breaking is unlikely if the water depth d at the toe of the wall is greater than 1.5 times the extreme wave height $H_{1/100}$, where:

$$H_{1/100} = 1.67\ H_s$$

In our study case:

$$H_{1/100} = 1.67 \cdot 5 = 8.35\text{ m}$$
$$1.5\ H_{1/100} = 1.5 \cdot 8.35 \approx 12.53\text{ m}$$

The water depth at the toe of the caisson is:

$$d\ = 16\text{ m}$$
$$16\text{ m} > 12.53\text{ m},$$

the waves at the structure are classified as non-breaking, and Sainflou's theory can be used.

Sainflou pressure distribution:

For Sainflou's method the design wave height is taken as:

$$H_{1/20} = 1.4\ H_s$$

The deep-water wavelength and wave number are:

$$L_0 = \frac{g\,T_p^2}{2\pi} \qquad k_0 = \frac{2\pi}{L_0}$$

The local wavelength at depth d is:

$$L\ = L_0 \tanh\left(\sinh\left(\sqrt{k_0 d}\right)\right) \qquad k = \frac{2\pi}{L}$$

The maximum water-surface elevation in front of the wall is:

$$\eta^* = H_{1/20} + \frac{\pi H_{1/20}^2}{L}\ \frac{1}{\tanh\,(kd)}$$

In our study case, η^*< hc so no overtopping occurs and the pressure diagram above MSL is triangular (no P2).

The pressure at mean sea level is:

$$p_1 = \gamma_w \left(d + \frac{H_{1/20}}{\cosh\,(kd)}\right) \frac{\eta^*}{d + \eta^*}$$

The pressure at the lowest point of the vertical wall (base of the trunk) is:

$$p_3 = \gamma_w \frac{H_{1/20}}{\cosh{(kd)}} \frac{h'}{d} - p_1 \left(\frac{h'}{d} - 1\right)$$

The pressure at the rear corner is taken equal to the front base pressure:

$$p_4 = p_3$$

Numerically, for our study case:

$$p_1 \approx 77.39 \text{ kN/m}^2$$
$$p_3 = p_4 \approx 55.01 \text{ kN/m}^2$$

Resulting horizontal and vertical forces (Sainflou)
From the pressure diagram, the horizontal forces on the caisson are:
On the submerged part (between base and MSL):

$$F_h = \frac{(p_1 + p_3)\, d}{2}$$

On the part between MSL and η^*:

$$F_{h1} = \frac{p_1\, \eta^*}{2}$$

The total horizontal wave force according to Sainflou is:

$$F_{h,\text{tot}} = F_h + F_{h1}$$

The resulting vertical force used in the sliding check is:

$$F_v = F_g - F_b - \frac{p_3\ 19}{2}$$

where F_g is the total weight of the caisson (already computed earlier), F_b is the buoyancy force, and the last term represents the vertical component of the dynamic pressure at the base.

The sliding safety check with Sainflou's forces is then:

$$\frac{\mu F v}{F h} > Cs$$

$$\frac{\mu F v}{F h} = 1.57 > C_s = 1.4$$

Since the calculated safety factor 1.571 is larger than the required value of 1.4, the vertical breakwater satisfies the sliding stability criterion according to Sainflou's method.

2) Goda's pressure distribution

For Goda's method, the design wave height is taken as:

$$H_{\max} = 1.8\, H_s$$

rather than $H_{1/20}$. The maximum water-surface elevation in front of the wall is:

$$\eta^* = 0.75(1 + cos\beta)Hmax$$

where β is the wave angle with respect to the breakwater. In our study case, normal wave attack is assumed:

$$\beta = 0^\circ$$

so that (1+\cos\beta) = 2.

The deep-water wavelength and the local wavelength at depth h are obtained from:

$$L_0 = \frac{\mathrm{g}\, T_p^2}{2\pi} \qquad k_0 = \frac{2\pi}{L_0}$$

$$L = L_0 \tanh\left(\sinh\left(\sqrt{k_0 d}\right)\right) \qquad k = \frac{2\pi}{L}$$

Goda coefficients and pressures

According to Goda, the pressure distribution on the seaward face of the vertical wall is defined through three coefficients α_1, α_2 and α_3:

$$\alpha_1 = 0.6 + \frac{1}{2}\left[\frac{4h\pi/L}{\sinh\,(4h\pi/L)}\right]^2$$

$$\alpha_2 = \min\left(\frac{h_b - d}{3h_b}\left(\frac{H_{\max}}{d}\right)^2,\ \frac{2d}{H_{\max}}\right)$$

$$\alpha_3 = 1 - \frac{h'}{h}\left(1 - \frac{1}{\cosh\,(2h\pi/L)}\right)$$

Where:

- h : is the total water depth,
- d: distance from still water level to caisson base (embankment height),
- h' : is the trunk height,
- H_b: height of the crown deck above the MSL

With these coefficients, the characteristic pressures become:

- Pressure at mean sea level:

$$P_1 = \frac{1}{2}(1 + \cos\ \beta)(\alpha_1 + \alpha_2 \cos^2 \beta)\gamma_w\, H_{\max}$$

- Pressure at the top of the wall (for $\eta^* > h_c$):

$$P_2 = P_1\, \frac{\eta^* - h_c}{\eta^*}$$

- Pressure at the base of the wall:

$$P_3 = \alpha_3\, P_1, \qquad P_4 = P_3$$

For our study case, the numerical results are:

$$p_1 \approx 76.85 \mathrm{kN/m}^2$$
$$p_2 \approx 19.92 \mathrm{kN/m}^2$$
$$p_3 = p_4 \approx 60.47 \mathrm{kN/m}^2$$

Resulting forces and sliding check (Goda)
From the pressure diagram, the horizontal forces on the caisson are:

- On the submerged part of the wall (base to MSL):

$$F_{h,\text{Goda}} = \frac{(p_1 + p_3)\, d}{2}$$

- On the part between MSL and the crown level h_{c}:

$$F_{h1,\text{Goda}} = \frac{(p_1 + p_2)\, h_c}{2}$$

The total horizontal wave force is:

$$F_{h,\text{tot,Goda}} = F_{h,\text{Goda}} + F_{h1,\text{Goda}}$$

The corresponding resulting vertical force used in sliding is:

$$F_{v,\text{Goda}} = F_g - F_b - \frac{p_3\ 19}{2}$$

where F_g is the total weight of the caisson and F_b is the buoyancy force (as previously computed).
The sliding safety factor according to Goda is:

$$\frac{\mu\, F_{v,\text{Goda}}}{F_{h,\text{tot,Goda}}} = 1.37$$

This value is smaller than the required sliding safety factor:

$$C_s = 1.4$$

According to Goda's theory, the vertical breakwater does not satisfy the sliding stability criterion, since the calculated safety factor 1.367 < 1.4.

2.6.2.3 Overturning check

Overturning stability
In addition to sliding, the overturning stability of the vertical breakwater has been checked under wave loading. The overturning analysis is carried out for the instant when a wave crest is acting on the caisson. In that situation, the wave-induced pressures on the seaward face generate a clockwise overturning moment around the landward toe (right-hand rotation in our sketch). If instead a wave trough were acting, the pressure distribution would be reversed and the caisson would tend to rotate in the opposite (left-hand) direction and around the seaward toe.

By summing the contributions of all overturning forces with respect to the landward toe, the overturning moment M_o is obtained. In our study case:

$$M_o \approx 5.07 \times 10^4 \text{ kNm}$$

calculated as:

$$\begin{aligned} M_o = \quad & (p_{2,\text{Goda}}\, h_c)(h' + 0.5h_c) + [(p_{1,\text{Goda}} - p_{2,\text{Goda}})\, 0.5h_c](h' + \frac{1}{3}h_c) \\ & +(p_{3,\text{Goda}}\, h')(0.5h') + [(p_{1,\text{Goda}} - p_{3,\text{Goda}})\, 0.5h'](\frac{2}{3}h') \\ & +(p_{4,\text{Goda}}\, 0.5 \cdot 19)(\frac{2}{3} \cdot 19) + F_b\, 9.5 \end{aligned}$$

The restoring moment M_r is mainly provided by the self-weight of the caisson acting through its lever arm with respect to the rotation point. In our case:

$$M_r = F_g \cdot 9.5 \approx 6.28 \times 10^4 \text{ kNm}$$

The overturning safety factor is then:

$$\frac{M_r}{M_o} = 1.24$$

The required safety factor against overturning is:

$$\left(\frac{M_r}{M_o}\right)_{\text{req}} = C_r = 1.5$$

Since

$$\frac{M_r}{M_o} = 1.24 < 1.5,$$

the vertical breakwater does not satisfy the overturning stability criterion under Goda’s wave loading.

The overturning stability of the vertical breakwater has also been evaluated using the Sainflou pressure distribution. As for the Goda case, the analysis is carried out for the instant when a wave crest is acting on the caisson, generating a clockwise overturning moment about the landward toe.

Starting from the Sainflou pressures (P_1, P_3, P_4), the corresponding resultant forces on the wall and at the base are determined, and their moments with respect to the landward toe are calculated. In addition, the buoyancy force on the base contributes to the overturning moment, while the self-weight of the caisson provides the resisting (restoring) moment.

In our study case the overturning moment according to Sainflou is computed as:

$$\begin{aligned} M_o = \quad & (p_{1,\text{Sainflou}} \cdot 0.5\,\eta^*)\left(h' + \frac{1}{3}\eta^*\right) \\ & + (p_{3,\text{Sainflou}} \cdot h')\,(0.5\,h') \\ & + [(p_{1,\text{Sainflou}} - p_{3,\text{Sainflou}})\,0.5\,h']\left(\frac{2}{3}h'\right) \\ & + (p_{4,\text{Sainflou}} \cdot 0.5 \cdot 19)\left(\frac{2}{3} \cdot 19\right) + F_b\,9.5 \end{aligned}$$

where η^* is the maximum water-surface elevation in front of the wall.
The restoring moment is:

$$M_r = F_g \cdot 9.5$$

Then you can close the Sainflou overturning section with the explicit numerical result:

$$\text{OT}_{\text{Sainflou}} = \frac{M_r}{M_o} \approx 1.35$$

Since

$$\text{OT}_{\text{Sainflou}} = 1.35 < 1.5,$$

the vertical breakwater does not satisfy the required overturning safety factor according to Sainflou's wave loading, even though the result is only slightly below the target value.

2.7 Extended Exercise

2.7.1 From analogue bathymetry to a digital depth dataset

The starting point of the exercise was an old bathymetric chart of the study area.
This analogue information first had to be converted into a digital depth dataset.

- The chart was imported into AutoCAD.
- All visible depth contours were traced as *polylines*, each polyline associated with a specific depth level z.
  Because the contours were already printed on the original map, they could simply be followed, ensuring that the new digital lines remained consistent with the historical data.

Once all depth contours were digitised, the drawing contained a set of 3D polylines: each point along a line had plan coordinates (x,y) and an elevation z (water depth).
This is effectively a vector representation of the bathymetry.
These polylines were then exported and processed in MATLAB to create a single table of bathymetric points, where every row corresponds to a point with its x-coordinate, y-coordinate, and water depth.
This table is the base bathymetry file used in the rest of the work.

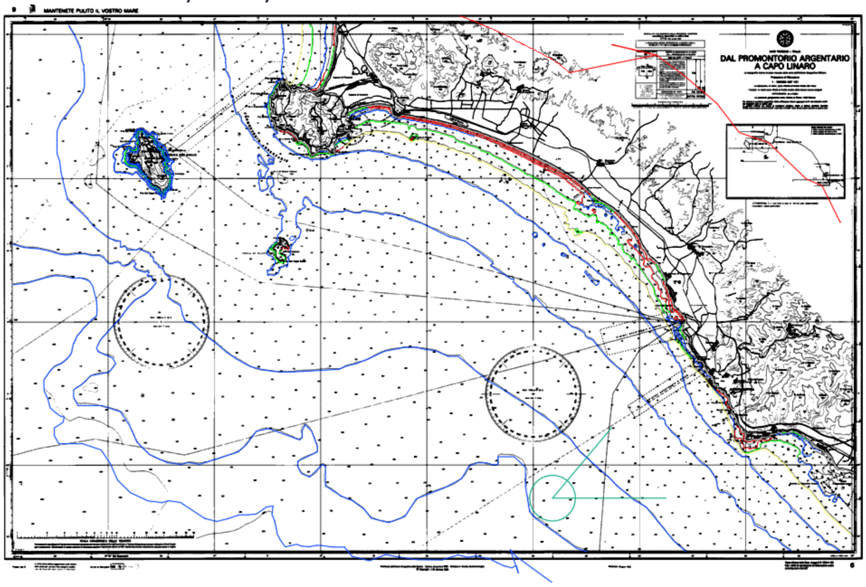

Figure 19: Chart in Autocad + polylines

2.7.2 Creation of the computational grid

With the scattered bathymetric points available, the next step was to construct a regular computational grid covering the area of interest.

- First, the model domain was defined: a rectangular area large enough to include the offshore region, the nearshore zone, and the coastline of interest.
  The size and position of this rectangle were chosen based on the geometry visible in the AutoCAD drawing.
- The grid was then oriented with respect to the coastline by applying a rotation. This alignment ensures that the grid axes are roughly parallel and perpendicular to the shore, which simplifies the interpretation of the results and later boundary condition definitions.
- Inside this rotated rectangle, a regular grid was created with a chosen spatial resolution. Each grid cell initially had only coordinates; no depth value.

The scattered bathymetric points obtained from the chart were then interpolated onto this grid. For every grid node, the depth was estimated based on the surrounding measured depths. The result is a gridded bathymetric surface: a matrix where each grid cell stores a depth value, representing a smoothed but coherent seabed for numerical modelling.

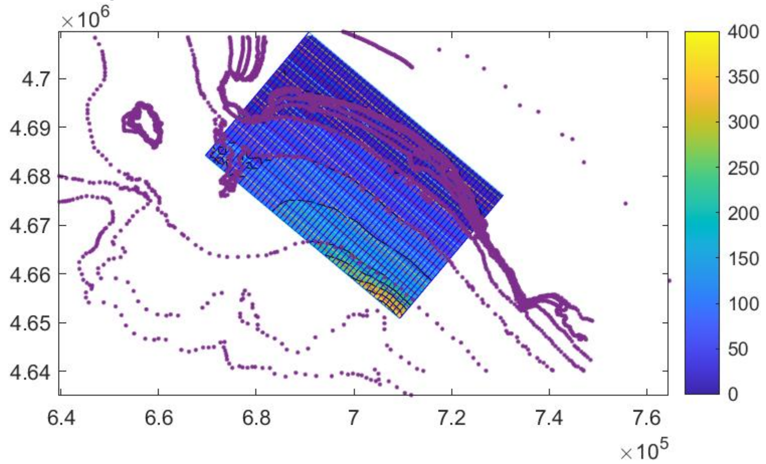

Figure 20: computational grid

The new depth grid was checked visually (for example with contour plots) to confirm that:

- the depth pattern followed the original bathymetric contours, and
- no unrealistic points were introduced by the interpolation.

This gridded bathymetry was then exported as a depth file and used as the seabed input for SWAN.

2.7.3 SWAN model set-up

With the bathymetric grid prepared, the next step was to set up the SWAN wave model.
In SWAN, the following elements were defined:

1. Computational mesh

The SWAN mesh was made consistent with the previously created grid:
same domain, same resolution and same coordinates.
This ensures that each SWAN grid point corresponds to one value in the depth file.

2. Bathymetry

The gridded depth file was supplied as the bottom topography.
SWAN uses this information to compute wave refraction, shoaling and depth-induced breaking.

3. Boundary conditions

In our SWAN set-up only one spectral boundary condition is applied, defined by:
BOUNDPAR2 SIDE SW CCW CON PAR 7 10 30 30
This corresponds to a parametric spectrum on the south-western side of the domain with:

- significant wave height: $H_s = 7m$
- peak period: $T_p = 10s$
- mean direction of wave propagation: 30°
- directional spreading: 30°

This single boundary represents one offshore wave system entering the domain from the south-west.
Because no wave conditions are prescribed on the other (lateral) boundaries, the model results near those open sides cannot be considered fully reliable. Reflections and numerical effects are not controlled there, so:
The SWAN results should only be trusted inside the domain away from the lateral boundaries, while values close to the unforced sides must be interpreted with caution.

4. SWAN results

The main SWAN output used in this exercise is the significant wave height field over the grid, together with the wave direction at each grid point, these results were visualized as follows:

- Grid plot
  We first plotted the grid lines of the computational domain to check that the model area and its rotation/orientation were correct.

- Map of significant wave height
  Next, we made a colour map of the significant wave height H_s over the whole domain.
  On top of this map we added arrows showing the local wave direction, so that we can see at the same time how high the waves are and from which direction they are coming.

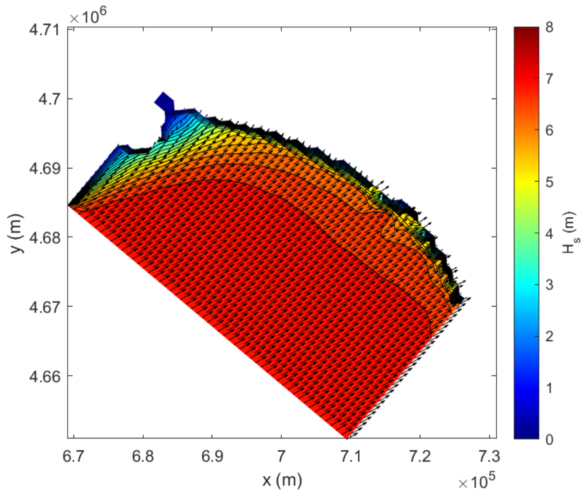

Figure 21: SWAN map of significant waveheight

- Transect analysis
  For our case, we selected row 11 of the grid as a cross-shore transect and extracted the values
  $h_{\text{transect}} = H_s(11,:), x_{\text{transect}} = x(11,:)$
  The significant wave height and the corresponding x-coordinates along that row.
  These data were plotted as H_s versus distance, giving a one-dimensional profile of wave height from offshore towards the coast.
  This simple plot clearly shows how the offshore wave height is transformed, decreasing due to shoaling, refraction and breaking as the waves move into shallower water near the shoreline.

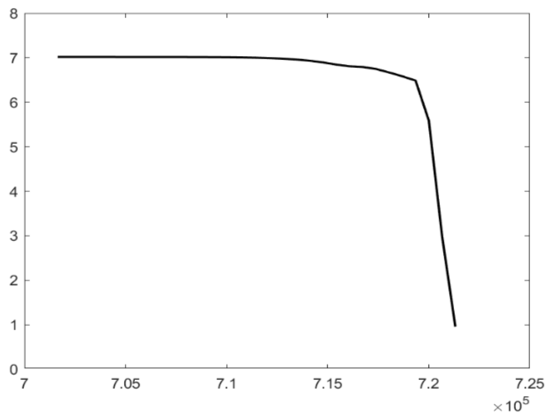

Figure 22: SWAN Transect analysis

3. Conclusion

In this report I have gone through the main steps that a coastal engineer must follow to design and check harbour structures. I started with short-term and long-term wave analysis to obtain reliable design wave parameters, using both zero-crossing and spectral methods and fitting an extreme-value distribution to long records. With linear wave theory I then studied how waves change when they travel from deep water towards the coast, including the effects of shoaling and refraction.

These wave conditions were used to design a rubble mound breakwater, where I checked wave breaking at the toe, sized the tetrapod armour and filter layers with Hudson-type formulas, and designed the toe berm using the Gerding approach. For the vertical breakwater I computed all relevant forces with and without waves, and performed sliding and overturning checks with Sainflou and Goda pressure theories. The Goda method turned out to be more demanding, and in my case the caisson did not fully satisfy the required safety factors, showing how sensitive vertical structures are to extreme wave loading.

In the extended exercise I learned how to move from an analogue bathymetric chart to a digital grid, build a numerical mesh and run a SWAN model. The model results showed how the offshore wave field transforms over the bathymetry and how wave heights change along a cross-shore transect. Overall, the course and this report helped me to connect theory with practice and to understand better how to analyze waves, design breakwaters and judge the reliability of numerical model results.